

Test0

Technical Mathematics Test Document

This document demonstrates and tests mathematical equations spanning both Algebra and Calculus, rendered using three distinct dialects: LaTeX, Typst typeset, and SymPy.

Skip to H3: Algebra Equations

Below are five algebra-themed equations.

1. **LaTeX (Algebra 1):** The expansion of a binomial squared.

$$(a + b)^2 = a^2 + 2ab + b^2$$

2. **LaTeX (Algebra 2):** The quadratic formula to solve $ax^2 + bx + c = 0$ in fraction format.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3. **SymPy (Algebra 3):** Factoring the difference of two squares.

$$a^2 - b^2 = (a - b)(a + b)$$

4. **Typst (Algebra 4):** The logarithmic product rule.

$$\log_{b(xy)} = \log_{b(x)} + \log_{b(y)}$$

5. **SymPy (Algebra 5):** The determinant of a 2×2 matrix.

$$ad - bc = ad - bc$$

Skip to H3: Calculus Equations

Below are five calculus-themed equations.

6. **LaTeX (Calculus 1):** The fundamental trigonometric limit.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

7. **Typst (Calculus 2):** The Fundamental Theorem of Calculus.

$$\int_a^b f(x) dx = F(b) - F(a)$$

8. **LaTeX (Calculus 3):** The derivative of the natural exponential function.

$$\frac{d}{dx}(e^x) = e^x$$

6. **Typst (Calculus 4):** The solution to the Basel problem.

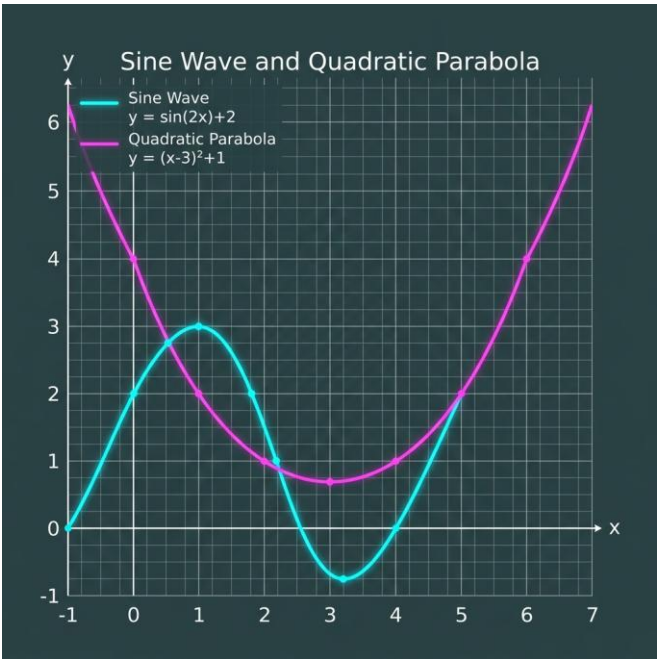
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

7. **SymPy (Calculus 5):** Derivative of the sine function.

$$\cos(x) = \cos(x)$$

Visual Representation

Here is a graph visualising some of the functions:



this image in this 49 is not alt text . put it

Tabular Data (Inaccessible Layout)

Value B			
Value A			
Row 1	12.5	14.2	19.1
Row 2	10.1	9.8	
Row 3	22.0		